

Quantum Wave Packet Scattering off a Schwarzschild Black Hole

A 3D Split-Operator Fourier Simulation Study

This report documents a fully three-dimensional numerical simulation of a non-relativistic quantum wave packet propagating in the presence of a Schwarzschild-inspired central potential. The wavefunction is evolved using a second-order split-operator Fourier method on a Cartesian grid, with complex absorbing boundary layers applied at every face of the simulation domain. Capture, reflection, and transmission probabilities are computed automatically by partitioning the computational domain relative to the event horizon and the initial direction of motion.

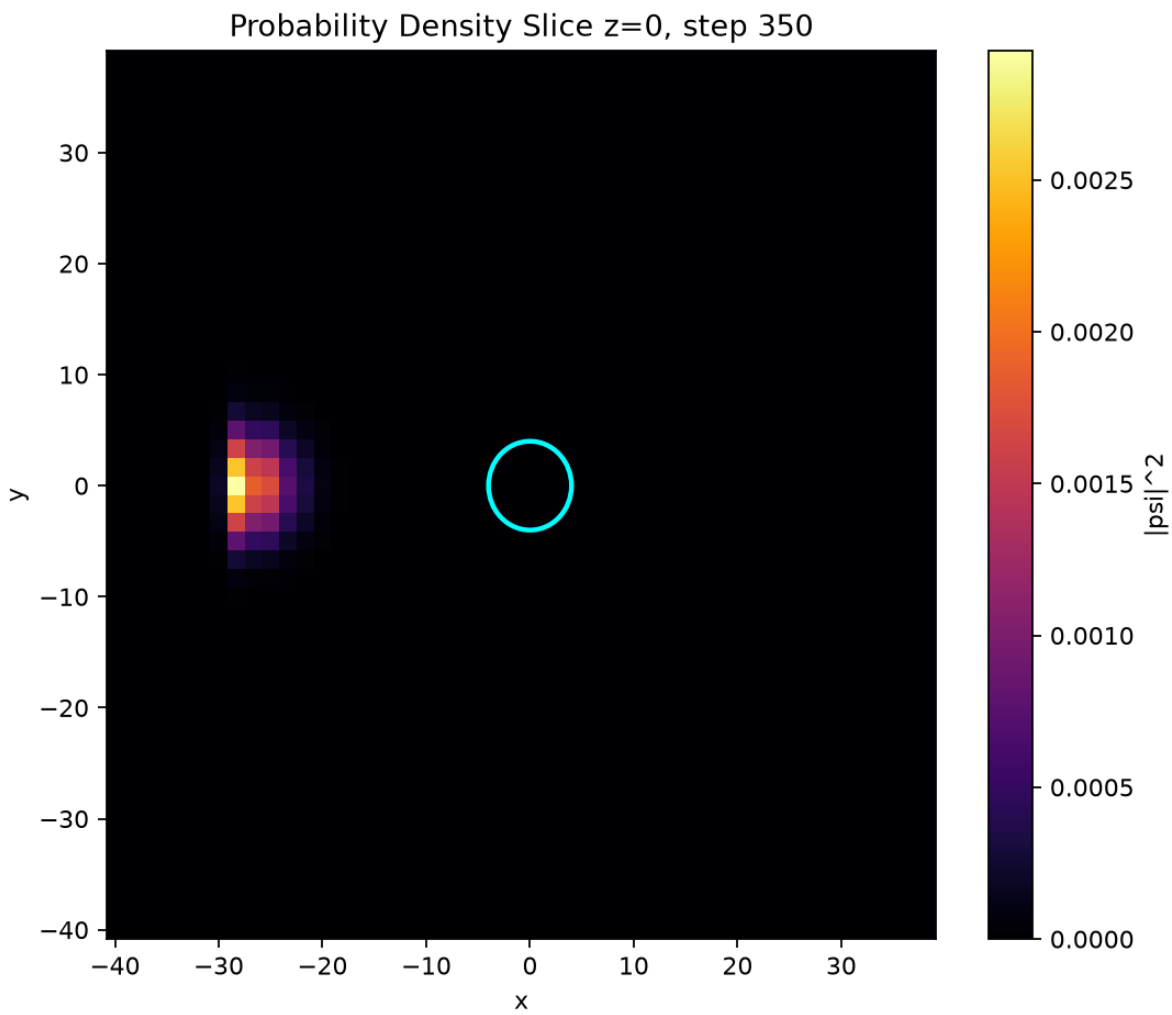
1. Simulation Configuration

Parameter	Value
Grid resolution	48 x 48 x 48
Domain size	80.0 x 80.0 x 80.0
Schwarzschild radius	4.0
Potential strength	1.0
Absorber width / strength	12.0 / 40.0
Initial packet center	(-25.0, 0.0, 0.0)
Initial momentum	(3.0, 0.0, 0.0)
Packet width (sigma)	3.0
Time step	0.01
Total steps	350

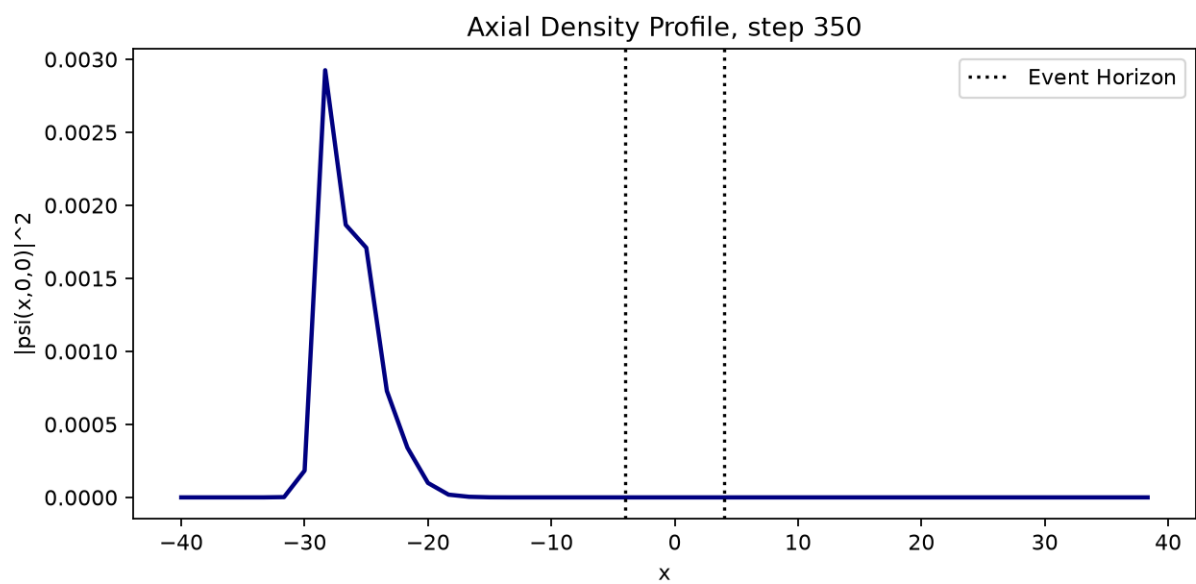
2. Baseline Run Results

Quantity	Value
time	3.500000
norm	0.769562
mean_x	-20.293614
mean_y	0.000000
mean_z	0.000000
mean_r	20.564404
kinetic_energy	0.259078
potential_energy	-0.034214
total_energy	0.224863
captured_probability	0.000001
reflected_probability	0.653358
transmitted_probability	0.116204

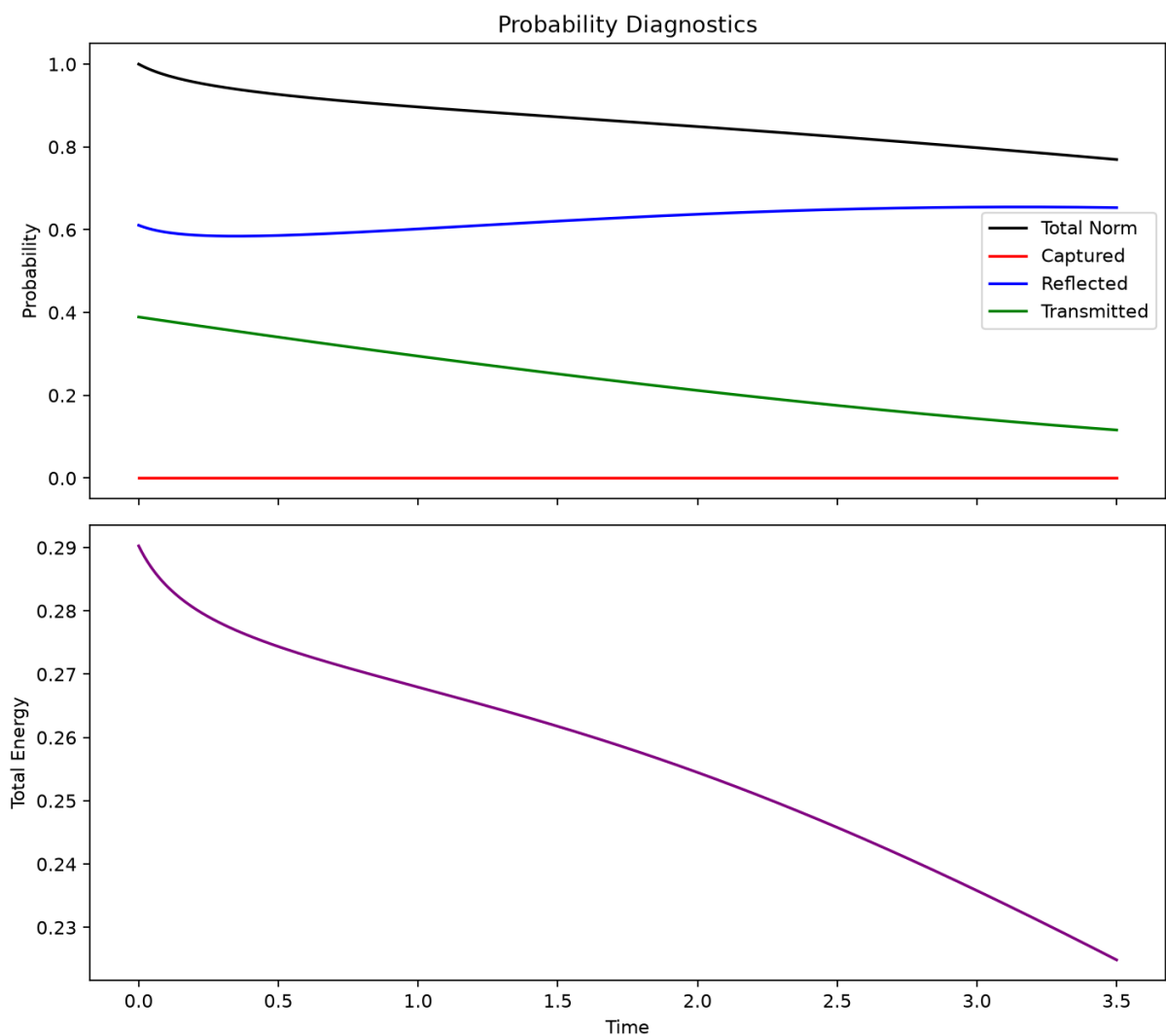
Probability Density Slice (z = 0)



Axial Density Profile

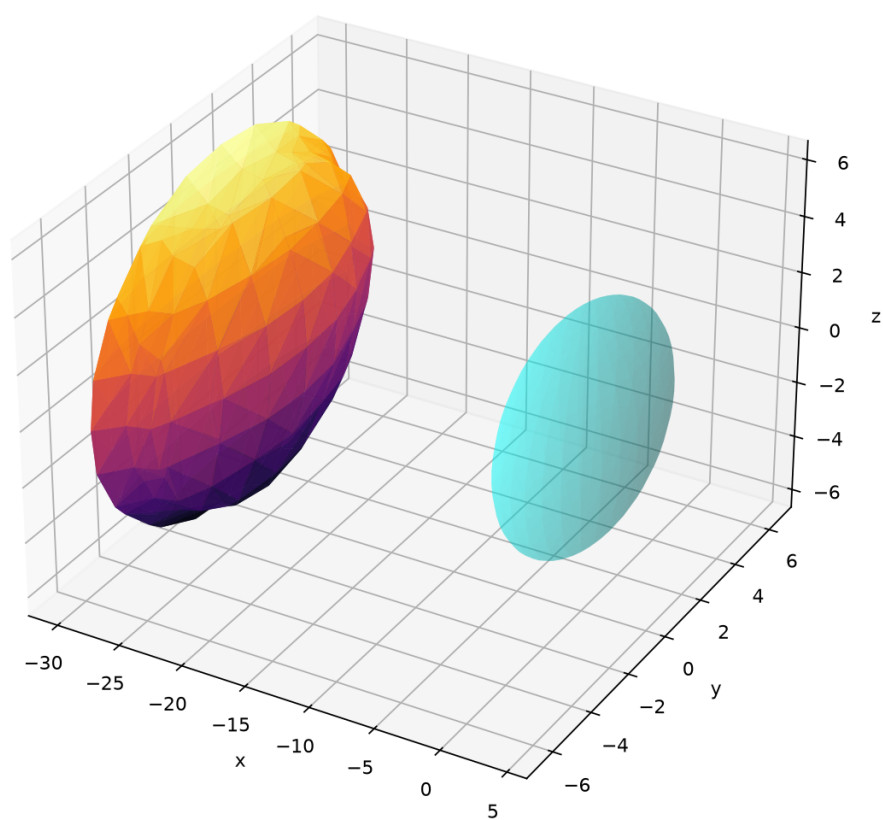


Probability and Energy Diagnostics



3D Probability Density Isosurface

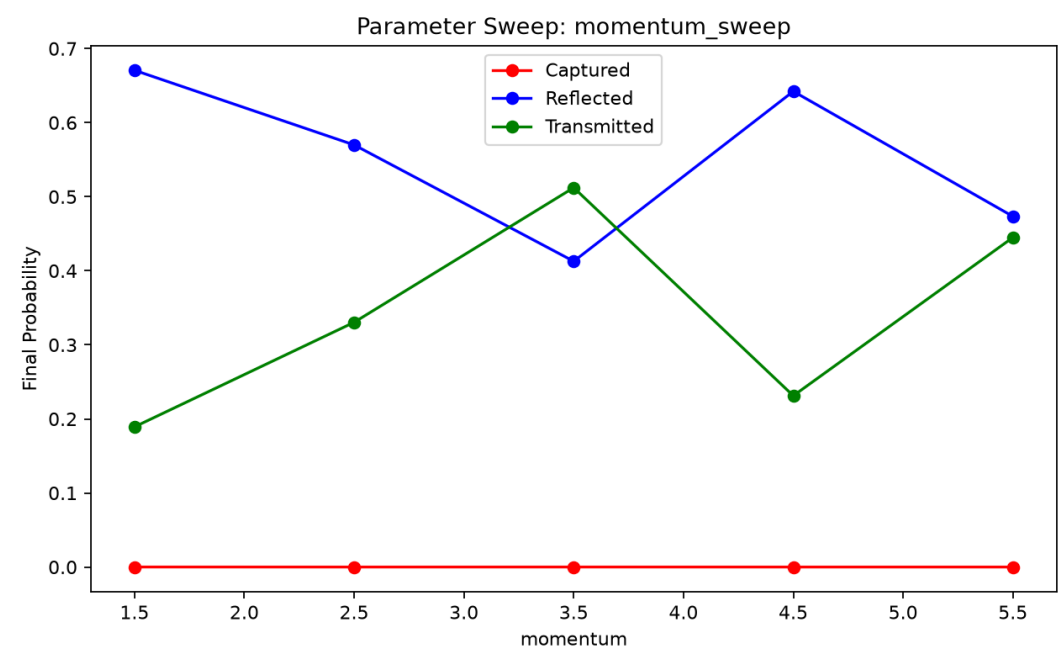
3D Probability Density Isosurface, step 350



3. Parameter Sweeps

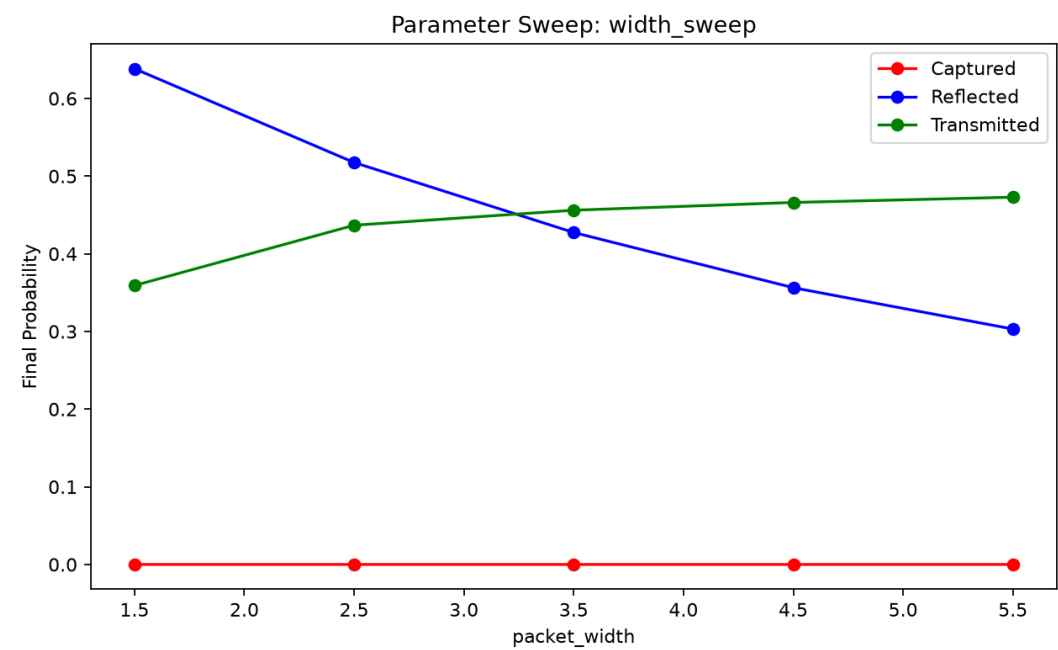
Momentum Sweep

Momentum	Norm	Captured	Reflected	Transmitted	Total Energy
1.5000	0.8600	0.0000	0.6707	0.1893	0.4069
2.5000	0.9004	0.0000	0.5699	0.3304	0.0081
3.5000	0.9250	0.0000	0.4128	0.5122	0.4235
4.5000	0.8735	0.0000	0.6422	0.2313	0.1250
5.5000	0.9184	0.0000	0.4734	0.4450	0.1079



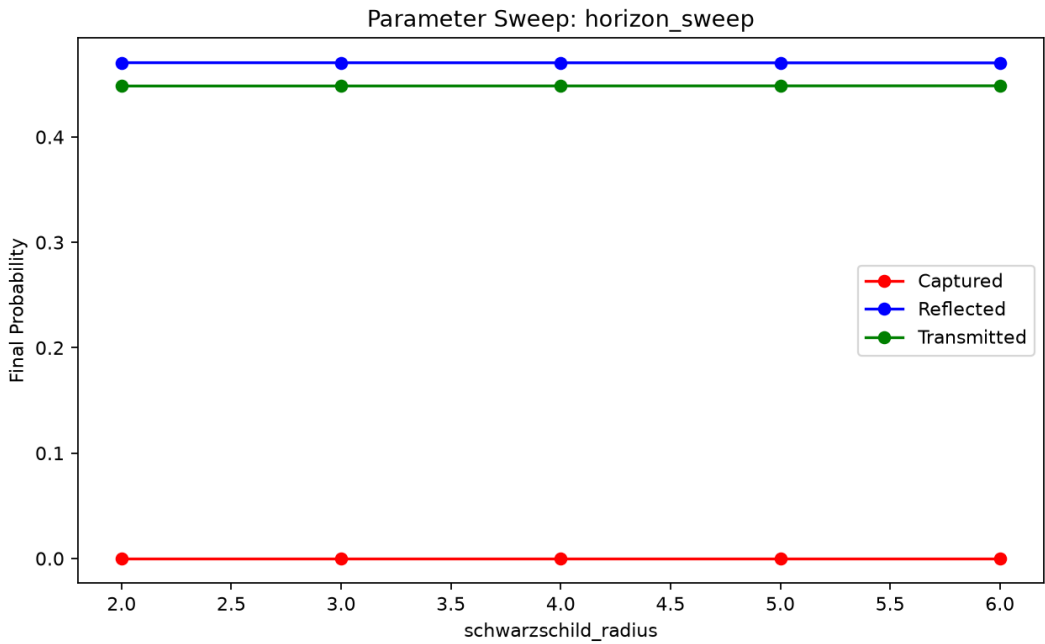
Width Sweep

Width	Norm	Captured	Reflected	Transmitted	Total Energy
1.5000	0.9975	0.0000	0.6383	0.3592	0.2382
2.5000	0.9543	0.0000	0.5175	0.4367	0.1343
3.5000	0.8835	0.0000	0.4274	0.4561	0.0997
4.5000	0.8225	0.0000	0.3564	0.4661	0.0819
5.5000	0.7760	0.0001	0.3031	0.4729	0.0708



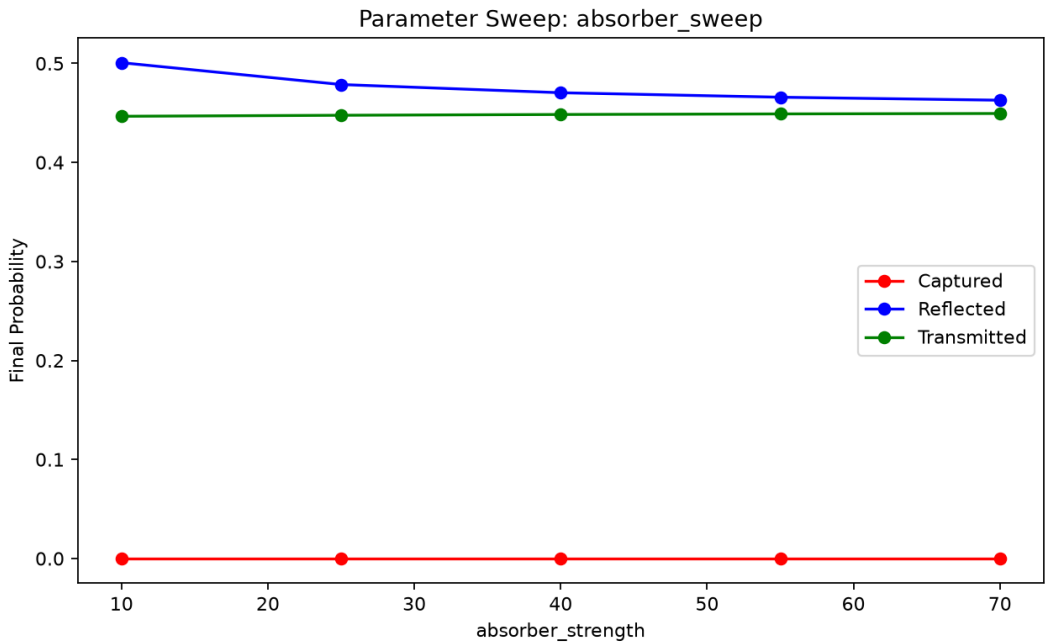
Horizon Sweep

Horizon Rs	Norm	Captured	Reflected	Transmitted	Total Energy
2.0000	0.9188	0.0000	0.4705	0.4483	0.1177
3.0000	0.9188	0.0000	0.4704	0.4484	0.1158
4.0000	0.9188	0.0000	0.4704	0.4484	0.1136
5.0000	0.9188	0.0000	0.4703	0.4485	0.1113
6.0000	0.9188	0.0000	0.4703	0.4485	0.1087



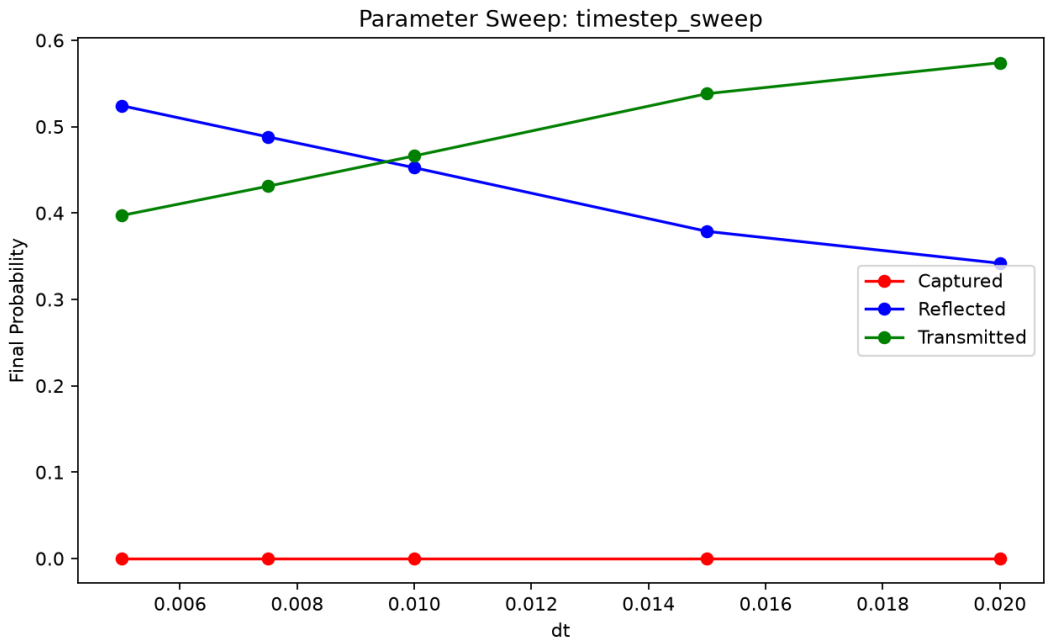
Absorber Sweep

Absorber	Norm	Captured	Reflected	Transmitted	Total Energy
10.0000	0.9473	0.0000	0.5007	0.4466	0.1109
25.0000	0.9263	0.0000	0.4787	0.4476	0.1122
40.0000	0.9188	0.0000	0.4704	0.4484	0.1136
55.0000	0.9148	0.0000	0.4658	0.4490	0.1144
70.0000	0.9122	0.0000	0.4628	0.4494	0.1148



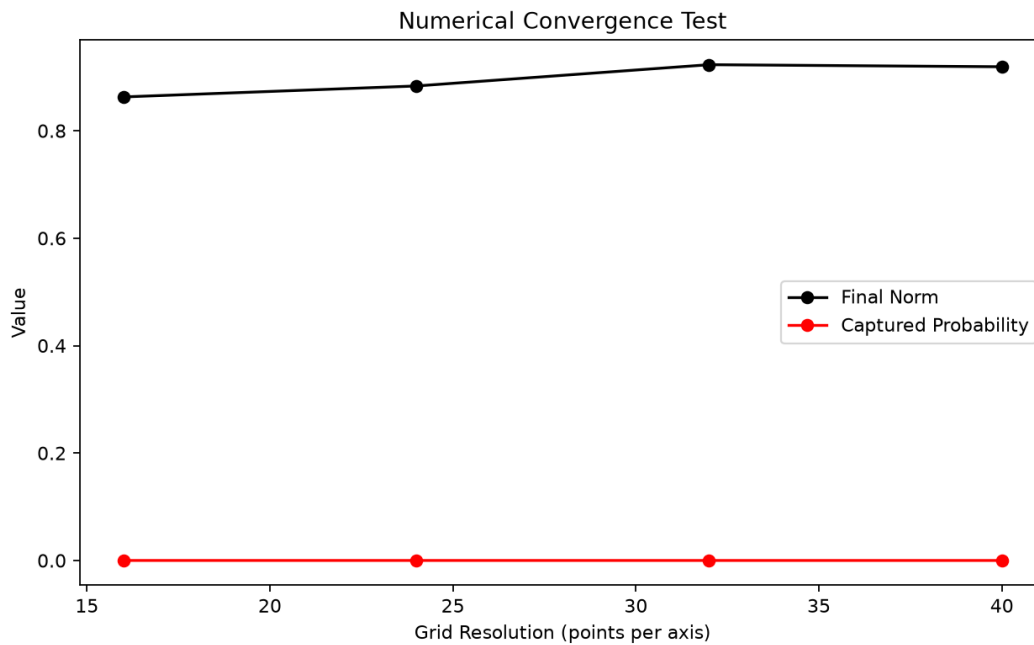
Timestep Sweep

Time Step	Norm	Captured	Reflected	Transmitted	Total Energy
0.0200	0.9158	0.0000	0.3418	0.5740	0.1146
0.0150	0.9168	0.0000	0.3787	0.5381	0.1146
0.0100	0.9185	0.0000	0.4524	0.4661	0.1141
0.0075	0.9191	0.0000	0.4881	0.4310	0.1130
0.0050	0.9214	0.0000	0.5243	0.3971	0.1114



4. Numerical Convergence Test

Resolution	Norm	Captured	Reflected	Transmitted	Total Energy
16	0.8629	0.0000	0.6761	0.1868	0.0688
24	0.8833	0.0000	0.4308	0.4525	0.2227
32	0.9229	0.0000	0.5344	0.3885	0.1109
40	0.9191	0.0000	0.4357	0.4834	0.0124



5. Summary

Under the baseline configuration, the simulation evolved a Gaussian wave packet with initial momentum (3.0, 0.0, 0.0) toward a Schwarzschild-inspired potential of horizon radius 4.0. The final captured probability was 0.000001, the reflected probability was 0.653358, and the transmitted probability was 0.116204. The parameter sweeps and resolution convergence test above characterize the sensitivity of these outcomes to the physical and numerical parameters of the model.